2. Hamiltonian Cycle Dry Run

Given Graph (Adjacency Matrix)

0 connected to 1,3

1 connected to 0,2,3,4

2 connected to 1,4

3 connected to 0,1,4

4 connected to 1,2,3

Step-by-step Execution

Step 1:

path = [0, -1, -1, -1, -1]

Step 2: pos = 1

Try vertex 1

path = [0,1,-1,-1,-1]

Step 3: pos = 2

Try vertex 2

path = [0,1,2,-1,-1]

Step 4: pos = 3

vertex 3 (no edge from 2→3)

vertex 4

path = [0,1,2,4,-1]

Step 5: pos = 4

vertex 3

path = [0,1,2,4,3]

Step 6: Check cycle

Is 3 connected to

0?

Final Output

Hamiltonian Cycle:

$0 \rightarrow 1 \rightarrow 2 \rightarrow 4 \rightarrow 3 \rightarrow 0$